STATEMENT OF WORK

RTL to GDS-II Implementation of

MIPI DSI Host Controller

for Foveated Rendering Systems

MTech VLSI Design — Physical Design Project

Project-Based Learning (PBL) — 6-8 Week Duration

Cadence EDA Suite: Genus | *Innovus* | *Tempus* | *Conformal* | *NCLaunch*

Team Size: 2 Members

Document Version: 1.0 — April 2026

1. Executive Summary

This Statement of Work defines the scope, methodology, deliverables, and timeline for the RTL-to-GDS-II implementation of a MIPI Display Serial Interface (DSI) Host Controller optimized for foveated rendering display systems. The project targets a complete ASIC physical design flow using the Cadence EDA suite available in the college laboratory.

The MIPI DSI protocol is the dominant display interface in mobile, automotive, and extended-reality (XR) devices, with over 80% of new mobile SoCs integrating DSI. Foveated rendering, which exploits the human eye's sharp acuity only in the central fovea (approximately 4% of display pixels), is becoming mandatory in VR/AR headsets to reduce rendering workload by 50–90%. The intersection of these two domains represents a research gap: no published academic work combines MIPI DSI host controller design with foveated-rendering-aware display pipeline optimization and takes it through a complete physical design flow.

The project will be executed in three phases over 6–8 weeks by a team of two, culminating in a synthesized, placed-and-routed, timing-clean GDS-II layout with comprehensive physical design reports on area, power, and timing.

2. Project Objectives

2.1 Primary Objectives

- Design and implement RTL (Verilog/SystemVerilog) for a MIPI DSI Host Controller supporting 1–4 data lane D-PHY interface with both Video Mode and Command Mode operation.
- Integrate a Foveated Region Mapper module that accepts gaze coordinates and dynamically maps display regions into foveal (full resolution), mid-peripheral (reduced quality), and outer-peripheral (minimum quality) tiers.
- Execute the complete RTL-to-GDS-II physical design flow using Cadence tools: functional simulation (NCLaunch/Xcelium), logic synthesis (Genus), formal verification (Conformal LEC), place-and-route (Innovus), and static timing analysis (Tempus).
- Achieve timing closure at a target clock frequency of 200 MHz or higher on a 45 nm standard cell library, with DRC and LVS clean GDS-II output.

2.2 Secondary Objectives

- Implement clock gating and operand isolation as low-power design techniques, reporting dynamic and leakage power.
- Write a basic UVM-based testbench for the D-PHY TX interface and packet layer, demonstrating constrained-random verification methodology.
- Document the complete design methodology in a format suitable for IEEE conference submission or journal publication.

3. Scope of Work

3.1 In Scope

The following items are explicitly within the scope of this project:

- **MIPI DSI Protocol Layer RTL:** Packet formation engine supporting short packets (4 bytes: DI + WC/Data + ECC) and long packets (4-byte header + payload up to 65,535 bytes + 2-byte CRC). ECC generator (Hamming 8,6), CRC-16 generator (polynomial $x^{16} + x^{12} + x^5 + 1$), virtual channel support (2-bit VC ID), and Display Command Set (DCS) command interface.
- **Lane Management Layer RTL:** 1/2/4 data lane distribution logic, byte-to-lane mapping with proper interleaving, and clock lane continuous/non-continuous clock modes.
- **D-PHY TX Digital Interface RTL:** High-Speed (HS) and Low-Power (LP) state machines implementing the D-PHY lane state transitions (Stop → HS-Rqst → HS-Prpr → HS-0 → HS-Data → HS-Trail → Stop), serializer for DDR data output, and LP escape mode entry sequences.
- **Foveated Region Mapper:** A configurable module that takes (gaze_x, gaze_y) coordinates as input and generates per-pixel or per-region quality tier signals. This module enables the controller to apply different DSI packet configurations (pixel format, compression ratio) based on the eccentricity from the gaze point.
- **Video Timing Controller:** Programmable horizontal and vertical sync, front/back porch, active pixel region generators supporting configurable resolutions up to 1920×1080.
- **APB Configuration Interface:** AMBA APB slave interface for register-level control of all controller parameters including resolution, pixel format, lane count, timing, foveation parameters, and mode selection.
- **Complete Physical Design Flow:** Synthesis with Genus (area, power, timing optimization), formal equivalence checking with Conformal LEC (RTL vs. gate-level netlist), floorplanning, placement, clock tree synthesis, routing with Innovus, and multi-corner multi-mode static timing analysis with Tempus.

3.2 Out of Scope

- Analog D-PHY transmitter circuit design (SLVS drivers, PLL, bias generators) — the project models the D-PHY at the digital PPI (PHY-Protocol Interface) boundary.
- DSI Receiver/Device controller RTL.
- Actual GPU rendering pipeline or eye-tracking hardware — gaze coordinates are provided as register-programmed or AXI-Stream inputs.
- Silicon fabrication or FPGA prototyping (though the RTL should be FPGA-friendly).
- VESA DSC/VDC-M compression codec implementation (noted as future work).

4. System Architecture

The MIPI DSI Host Controller architecture consists of seven major RTL modules organized in a datapath pipeline from the video input interface through to the D-PHY PPI output. The design operates across two primary clock domains: a pixel clock domain (for video input processing) and a byte clock domain (for DSI protocol and D-PHY serialization), connected via an asynchronous FIFO.

4.1 Module Hierarchy

Module	Key Functions	Interfaces
Video Timing Controller	H/V sync generation, blanking, active pixel windowing	DPI input (RGB + sync + DE) → Pixel FIFO
Foveated Region Mapper	Gaze-to-region mapping, quality tier assignment (3 tiers)	(gaze_x, gaze_y) input → tier[1:0] per pixel/region
Pixel-to-Byte Packer	RGB888/666/565 → byte stream, pixel format encoding	Pixel FIFO → Byte stream + format ID
Packetizer (ECC/CRC)	Short/long packet formation, Hamming ECC, CRC-16	Byte stream → DSI packets with headers/footers
Lane Manager	1/2/4 lane distribution, byte interleaving	DSI packets → Per-lane byte streams
D-PHY TX Controller	HS/LP state machines, serializer, clock lane FSM	Per-lane bytes → PPI signals (TxDataHS, TxRequestHS, etc.)
APB Register File	Configuration registers, status, interrupt	APB slave interface (PADDR, PWDATA, PRDATA, etc.)

4.2 Foveated Rendering Integration

The Foveated Region Mapper is the novel architectural contribution of this project. It implements a concentric-ring model based on the eccentricity from the current gaze fixation point. The mapper divides the display into three quality tiers:

- **Tier 0 (Foveal, 0°–5° eccentricity):** Full RGB888 resolution, no quality reduction. Covers approximately 4–8% of display area. DSI packets carry uncompressed 24-bpp pixel data.
- **Tier 1 (Mid-Peripheral, 5°–20°):** Reduced to RGB666 (18-bpp) or spatially subsampled. Covers approximately 20–30% of display area. Achieves 25% bandwidth reduction per pixel.

- **Tier 2 (Outer-Peripheral, >20°):** Reduced to RGB565 (16-bpp) with optional 2×2 spatial downsampling. Covers 60–75% of display area. Achieves 33–67% bandwidth reduction per pixel.

The net effect is a significant reduction in DSI link bandwidth utilization (estimated 40–60% reduction over uniform full-resolution streaming), directly translating to lower D-PHY power consumption and enabling higher refresh rates within the same lane bandwidth budget. This bandwidth savings is the key physical-design-relevant contribution, as it directly impacts dynamic power in the serializer and clock distribution network.

5. Design Methodology and Tool Flow

The project follows a standard industry RTL-to-GDS-II ASIC design methodology adapted for the Cadence EDA suite. Each tool step produces verified outputs that feed the next stage, with formal equivalence checks at critical handoff points.

5.1 RTL Design and Functional Verification

Tool: NCLaunch / Xcelium Simulator

- RTL coded in synthesizable Verilog-2001 / SystemVerilog, following coding guidelines for synthesis (no latches, complete sensitivity lists, synchronous resets).
- Directed testbenches for each module with self-checking assertions (SVA) for protocol compliance: ECC correctness, CRC validation, HS/LP state machine sequences, packet format compliance.
- A top-level integration testbench driving a 1080p/60fps video pattern through the complete pipeline and verifying output DSI packet sequences.
- Optional: Basic UVM testbench for D-PHY TX with constrained-random stimulus, functional coverage, and scoreboard.
- **Key deliverable:** Simulation waveforms, functional coverage report, lint-clean RTL.

5.2 Logic Synthesis

Tool: Cadence Genus Synthesis Solution

- **Target library:** 45 nm standard cell library (e.g., NanGate 45nm Open Cell Library or TSMC 45nm educational PDK, depending on college license).
- **Constraints (SDC):** Clock definitions for pixel_clk (~148.5 MHz for 1080p/60) and byte_clk (~200 MHz), input/output delays, false paths across clock domain crossing FIFO, max transition, max fanout, and clock uncertainty.
- Synthesis flow: syn_generic → syn_map → syn_opt with area and power optimization directives.
- Clock gating insertion enabled (set_attribute lp_insert_clock_gating true) for automatic clock gating of register banks.

- **Key deliverables:** Gate-level netlist (.v), SDC constraints, area report, power report, timing report (setup/hold), QoR summary.

5.3 Formal Equivalence Checking

Tool: Cadence Conformal Logic Equivalence Checker (LEC)

- RTL-to-gate-level netlist equivalence verification at two checkpoints: post-synthesis (RTL vs. Genus netlist) and post-place-and-route (RTL vs. Innovus netlist).
- Setup: read_library, read_design (golden = RTL, revised = netlist), set_system_mode lec, add_compared_points, compare.
- **Key deliverable:** LEC pass/fail report with 100% equivalence on all compared points.

5.4 Physical Design (Place and Route)

Tool: Cadence Innovus Implementation System

The physical design flow consists of the following sequential steps:

- **Floorplanning:** Define die area and aspect ratio, place I/O pins for APB interface and D-PHY PPI outputs, create power ring and power stripes (VDD/VSS), define placement blockages for analog macro placeholders.
- **Power Planning:** Global power grid with horizontal VDD/VSS stripes, power ring around core, IR drop analysis to ensure <5% voltage drop.
- **Placement:** Standard cell placement with timing-driven and congestion-aware optimization. Placement of clock-domain-crossing FIFO near the boundary between pixel and byte clock regions.
- **Clock Tree Synthesis (CTS):** Build balanced clock trees for pixel_clk and byte_clk domains with target skew <100 ps. Clock tree buffers/inverters, useful skew optimization, and post-CTS timing optimization.
- **Routing:** Global routing followed by detailed routing with DRC clean output. Signal integrity analysis for crosstalk on critical timing paths.
- **Optimization:** Post-route timing optimization (optDesign -postRoute), hold fixing, and area recovery.
- **Key deliverables:** Floorplan DEF, placed DEF, routed DEF, GDS-II, final netlist, SPEF parasitics, layout screenshots, DRC/LVS clean reports.

5.5 Static Timing Analysis (Signoff)

Tool: Cadence Tempus Timing Signoff Solution

- Multi-corner multi-mode (MCMM) analysis: at minimum, worst-case slow (setup), best-case fast (hold), and typical corners.
- Clock domain crossing (CDC) paths between pixel_clk and byte_clk domains verified as false paths or with proper synchronizer constraints.

- Reporting: worst negative slack (WNS), total negative slack (TNS), worst hold slack, and per-clock-domain timing summaries.
- **Key deliverable:** Timing signoff report showing zero setup/hold violations across all corners, or documented waivers for any remaining violations with root cause analysis.

5.6 Power Analysis

Tool: Cadence Voltus (if available) or Genus/Innovus power reporting

- Switching activity annotation from simulation VCD files for accurate dynamic power estimation.
- Leakage power analysis at worst-case temperature corner.
- Comparison of power with foveated rendering enabled (Tier 0/1/2 mixed) versus uniform full-resolution mode to quantify the power savings from foveation.
- **Key deliverable:** Power report showing total, dynamic, and leakage power breakdown per module, plus foveated vs. non-foveated comparison.

6. Project Timeline

The project spans 8 weeks divided into three phases aligned with the presentation schedule. Phase 1 concludes at week 2 with the first presentation, Phase 2 runs through week 5, and Phase 3 (final implementation and signoff) concludes at week 6–8.

6.1 Phase 1: Foundation and RTL Design (Weeks 1–2)

Milestone: Phase 1 Presentation

Week	Activities	Deliverables	Owner
Week 1	Literature survey completion Study DSI v1.01 spec: packet structure, ECC/CRC, lane management Study D-PHY v1.1 spec: HS/LP state machines, PPI interface Study foveated rendering fundamentals (Wang 2023 survey, Patney 2016) Analyze open-source DSI controllers (skatanik, briansune, zoatel repos) Define micro-architecture specification document	Literature survey report Architecture spec (block diagram, interfaces, register map) SDC constraints draft	Both members
Week 2	RTL coding: Video Timing Controller, Pixel-to-Byte Packer	RTL source files (4–5 modules) Simulation waveforms	Split: Member A (Protocol), Member B (PHY + Foveation)

	RTL coding: ECC generator, CRC-16 generator, Packetizer RTL coding: Foveated Region Mapper (parameterized ring model) Directed testbenches for each module Phase 1 presentation preparation	Phase 1 presentation slides	

6.2 Phase 2: RTL Completion, Verification, and Synthesis (Weeks 3–5)

Week	Activities	Deliverables	Owner
Week 3	RTL coding: Lane Manager (1/2/4 lane modes) RTL coding: D-PHY TX Controller (HS/LP FSMs, serializer, clock lane) RTL coding: APB Register File Top-level integration and CDC FIFO	Complete RTL hierarchy Top-level integration testbench	Split: Member A (Lane Mgr + APB), Member B (D-PHY TX + Integration)
Week 4	Full-chip functional verification with 1080p test patterns SVA assertions for protocol compliance Lint cleanup (HAL, SpyGlass rules or Genus lint) Begin Genus synthesis: initial run, constraint debug, timing analysis	Simulation pass with all test cases Lint-clean RTL Initial synthesis reports (area, power, timing)	Both members collaborating
Week 5	Synthesis optimization: iterate on constraints, fix timing violations, optimize area Conformal LEC: RTL vs. synthesized netlist Gate-level simulation with SDF back-annotation (optional but recommended) Begin Innovus: import design, floorplanning, power planning	Optimized gate-level netlist Conformal LEC pass report Floorplan and power grid	Split: Member A (Synthesis + LEC), Member B (Innovus floorplan)

6.3 Phase 3: Physical Design and Signoff (Weeks 6–8)

Milestone: Phase 2 Presentation (Week 6) + Final Presentation (Week 8)

Week	Activities	Deliverables	Owner
Week 6	Innovus: Placement, CTS (both clock domains), post-CTS optimization Innovus: Global and detailed routing, DRC fixing Post-route timing analysis with extracted parasitics (SPEF) Phase 2 presentation preparation	Placed and routed DEF Post-route timing report Phase 2 presentation slides	Both members
Week 7	Tempus signoff STA: MCMM analysis (slow/fast/typical corners) Hold fixing and ECO iterations Conformal LEC: RTL vs. final post-route netlist Power analysis with VCD-based switching activity GDS-II export from Innovus	Timing-clean STA report Final LEC pass report Power analysis report GDS-II file	Split: Member A (STA + Power), Member B (LEC + GDS)
Week 8	Final report writing: complete documentation of methodology, results, analysis Comparative analysis: foveated vs. non-foveated power/bandwidth Final presentation preparation with demo (layout walkthrough, timing report)	Final project report Final presentation slides Complete design database	Both members

7. Deliverables Summary

Phase 1 (Week 2)	Phase 2 (Week 6)	Final (Week 8)
Literature survey report Architecture specification RTL: 4–5 core modules Module-level simulations SDC constraints (draft) Presentation slides	Complete RTL hierarchy Full-chip simulation pass Genus synthesis reports Conformal LEC (post-syn) Innovus floorplan + P&R Post-route timing report Presentation slides	GDS-II layout (DRC clean) Tempus STA (MCMM signoff) Conformal LEC (post-route) Power analysis report Foveated vs. uniform comparison Final project report Final presentation

8. Risk Assessment and Mitigation

Risk	Severity	Impact	Mitigation
Timing closure failure at 200 MHz	High	Cannot demonstrate target frequency	Start with conservative 100 MHz target; incrementally increase. Use multi-Vt cells.
PDK/library unavailability	High	Cannot run synthesis/PnR	Verify library access in Week 1. Fallback: use NanGate 45nm Open Cell Library (free).
RTL complexity exceeds time budget	Medium	Incomplete RTL delays synthesis	Prioritize core DSI pipeline; APB and foveation mapper are simplifiable. Reuse open-source ECC/CRC modules.
CDC (clock domain crossing) issues	Medium	Metastability, data corruption	Use proven async FIFO design (Gray-code pointer). Constrain as false paths in SDC.
Conformal LEC non-equivalence	Medium	Cannot verify functional correctness	Run LEC incrementally after each optimization. Use guided setup for scan-inserted netlists.
DRC violations in routing	Low	GDS-II not clean for signoff	Use conservative utilization (60–70%). Run DRC after each routing iteration.

9. Expected Results and Success Criteria

Based on comparable published RTL-to-GDS-II implementations of protocol controllers (AHB-Lite, UART, RISC-V cores) at 45 nm technology, the following results are anticipated:

Metric	Target	Stretch Goal
Operating Frequency	≥ 200 MHz byte clock	≥ 300 MHz (enabling 2.4 Gbps/lane)
Gate Count	40K–80K gates	< 50K gates (area optimized)
Total Power	< 50 mW at 200 MHz	< 30 mW with aggressive clock gating
Core Area	< 0.5 mm² at 45 nm	< 0.3 mm²
Foveated Bandwidth Savings	≥ 40% DSI link utilization reduction	≥ 60% with 3-tier foveation
Timing Closure	Zero setup violations (WNS ≥ 0)	Positive slack on all paths

DRC/LVS	Clean (zero violations)	Clean

10. Industry Relevance and Career Impact

10.1 Relevance to Qualcomm and Semiconductor Industry

This project is exceptionally well-aligned with Qualcomm's core business and active hiring areas. Qualcomm is a Promoter member of the MIPI Alliance and Vice-Chairs the MIPI Display Working Group. Every Snapdragon SoC integrates a MIPI DSI host controller as its primary display interface through the Adreno Display Processing Unit (DPU). Qualcomm also pioneered hardware foveated rendering with the Snapdragon 835's Adreno Foveation and continues to advance it in the XR2 Gen 2 platform powering Meta Quest 3 and Samsung XR devices.

10.2 Skills Demonstrated for Physical Design (PD) Roles

- **End-to-End RTL-to-GDS-II Flow Competency:** The most directly assessed skill in PD interviews. You will demonstrate synthesis constraint writing (SDC), floorplanning, CTS, timing closure, and signoff.
- **Multi-Clock Domain Design:** CDC handling between pixel and byte clock domains is a frequently tested interview topic. Your async FIFO and false-path constraints demonstrate this directly.
- **Protocol-Level Understanding:** MIPI DSI is one of the key protocols asked about in Qualcomm and Samsung interviews for display-related positions.
- **Low-Power Design Techniques:** Clock gating, power gating awareness, and dynamic power optimization are core PD interview topics.
- **EDA Tool Scripting (TCL):** Every PD role requires TCL scripting for tool automation. Your Genus/Innovus/Tempus scripts become portfolio artifacts.

10.3 Skills Demonstrated for Design Verification (DV) Roles

- **UVM Methodology:** Even a basic UVM testbench for the D-PHY TX demonstrates agent architecture, constrained-random verification, functional coverage, and scoreboard development.
- **Protocol Verification:** SVA assertions for DSI packet compliance, ECC/CRC checking, and state machine coverage are directly applicable to DV roles at Qualcomm, Intel, and Samsung.
- **Formal Verification:** Conformal LEC experience demonstrates understanding of equivalence checking, a growing area in DV.

10.4 Resume Positioning

For a resume or interview, this project should be positioned as: "Designed and implemented a MIPI DSI v1.3 Host Controller with novel foveated-rendering-aware bandwidth optimization, achieving complete RTL-to-GDS-II signoff at 45 nm using Cadence Genus, Innovus, Tempus, and Conformal. Demonstrated 40–60% DSI link bandwidth reduction through 3-tier gaze-contingent pixel format adaptation, with timing closure at 200 MHz and power under 50 mW."

11. Key References

11.1 Specifications

- MIPI Alliance, "MIPI DSI Specification v1.01.00," February 2008.
- MIPI Alliance, "MIPI D-PHY Specification v1.1."
- MIPI Alliance, "MIPI DSI-2 Specification v2.2."

11.2 Key Academic Papers

- V. Melikyan et al., "A UVM-based Verification Approach for MIPI DSI Low-Level Protocol Layer," IEEE 2022.
- B.D. Kim et al., "Design of D-PHY chip for mobile display interface supporting MIPI standard," Microelectronics Journal, 2012.
- K. Kwon et al., "Low-Cost Unified Pixel Converter from MIPI DSI Packets," MDPI Electronics, 2022.
- H. Wang et al., "POLO: Process Only Where You Look," ISCA 2025.
- L. Wang et al., "Foveated Rendering: A State-of-the-Art Survey," Computational Visual Media, 2023.
- A. Patney et al., "Towards Foveated Rendering for Gaze-Tracked VR," ACM TOG/SIGGRAPH Asia, 2016.
- S. Hiremath et al., "RTL to GDSII Implementation of AHB-Lite," IEEE ICCES, 2021.

11.3 Open-Source Repositories

- skatanik/dsi_controller — Complete MIPI DSI controller in Verilog (github.com/skatanik/dsi_controller).
- briansune/Kintex-7-MIPI-DSI — Modular pure-Verilog DSI TX (github.com/briansune/Kintex-7-MIPI-DSI-5.5-inch-LCD).
- zoatel/MIPI-D-PHY — D-PHY TX RTL + UVM testbench (github.com/zoatel/MIPI-D-PHY).
- TILOS-AI-Institute/MacroPlacement — Cadence Genus + Innovus flow scripts (github.com/TILOS-AI-Institute/MacroPlacement).
- abdelazeem201/Cadence-RTL-to-GDSII-Flow — Cadence flow tutorial (github.com/abdelazeem201/Cadence-RTL-to-GDSII-Flow).